

Figure 1: Desktop operating systems' market share

org) and for Notepad++ (<https://notepad-plus-plus.org>). Now, I would like to consider an imaginary programming language called *qwerty*. In the *qwerty* language, a comment line starts with a double slash (`//` this is a comment) and the strings are those between two quotation marks ("`this is a string`"). This language has many keywords which can be classified into various categories. In jEdit, the syntax highlighting file has the following structure:

```
<?xml version="1.0"?>
<!DOCTYPE MODE SYSTEM "xmode.dtd">
<!-- jedit syntax file for the qwerty language -->
<MODE>
<RULES IGNORE_CASE="FALSE" HIGHLIGHT_DIGITS="FALSE">
<SPAN TYPE="LITERAL1" ESCAPE="\\" NO_LINE_BREAK="FALSE">
<BEGIN"/><BEGIN>
<END"/><END>
</SPAN>
<EOL_SPAN TYPE="COMMENT1"/></EOL_SPAN>
<KEYWORDS>
<!-- Category 1 -->
<KEYWORD1>this-is-a-keyword</KEYWORD1>
...
<!-- Category 2 -->
<KEYWORD2>this-is-a-keyword</KEYWORD2>
...
<!-- Category 3 -->
<KEYWORD3>this-is-a-keyword</KEYWORD3>
</KEYWORDS>
</RULES>
</MODE>
```

Because I'm not interested in highlighting the digits, this option is defined as 'false'. Save the file as, for example, *qwerty.xml*, and put it in the *modes* directory (`C:\Program Files\jEdit\modes`). Then add the following line to the catalogue file located in the same directory:

```
<MODE NAME="qwerty" FILE="qwerty.xml" FILE_NAME_GLOB="*.qwe"
/>
```

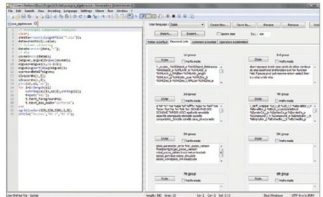


Figure 2: Notepad++ custom language editor

For Scilab (<http://www.scilab.org>), the complete file about this first example is freely (as in 'freedom' and in 'beer') available at <https://github.com/astonfe/scilab>. In Notepad++, a syntax highlighting file is not very different from that for jEdit, and has a structure divided more or less into three sections. The comment line, math operators, punctuation, parentheses and some keywords, for example, in an if-else-end statement are in the first section:

```
<Keywords name="Comments">%%/</Keywords>
<Keywords name="Operators1">+ * / = &apos; , ; . : &lt; &gt; &lt; &gt; ( ) [ ] { } | &amp; ; ~</Keywords>
<Keywords name="Folders in code2, open">for function if
select try while</Keywords>
<Keywords name="Folders in code2, middle">case catch else
elseif</Keywords>
<Keywords name="Folders in code2, close">end endfunction</
Keywords>
```

The keywords according to the following simple schema are defined in the second section:

```
<Keywords name="Keywords1">
this-is-a-keyword
...
</Keywords>
<Keywords name="Keywords2">
this-is-a-keyword
...
</Keywords>
<Keywords name="Keywords3">
this-is-a-keyword
...
</Keywords>
```

The strings, as well as the colours and fonts for each of the above categories, are defined in the last section. Save as *UserDefineLang.xml* and copy to `C:\Users\AppData\Roaming\Notepad++`. This editor also has its own built-in editor for custom languages, so it's not necessary to directly